Quantification: 536-63x_Hyp_RD_z08c1-2
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PRE-VALIDATION: no number here has been checked against manual (ImageJ) counts.
PIXEL UNITS: lengths px, areas px^2, densities px^-1. The images carry no micron calibration.
Statistics are over interior lacunae only (those not touching the frame edge).

Per image
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measure	value
lacuna count	29
interior lacuna count	25
frame-edge lacuna count	4

Interior lacunae only: mean, median and sample SD (n = 25)
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measure	kind	mean	median	sd	units
area_px2	lacuna shape	2617	2458.5	1727.14	px^2
major_axis_length_px	lacuna shape	92.4973	99.3629	42.6278	px
minor_axis_length_px	lacuna shape	39.7853	38.7847	14.5046	px
aspect_ratio	lacuna shape	2.3245	1.9781	0.8851	unitless
eccentricity	lacuna shape	0.8543	0.8626	0.1031	unitless
solidity	lacuna shape	0.7729	0.7971	0.1213	unitless
roots_count	headline	9.32	8	5.7498	count
ring_length_r30_px	headline	279.48	261	162.985	px
ring_length_r60_px	reported	550.04	498	329.151	px
owned_length_px	ownership-dependent	733.804	617.286	569.392	px
edge_count	ownership-dependent	35.28	31	27.8546	count
mean_edge_length_px	ownership-dependent	21.7581	21.5278	6.5576	px
perimeter_px	normalised	290.831	259.557	157.916	px
ring_area_r30_px2	normalised	8361.24	9187	3584.47	px^2
ring_area_r60_px2	normalised	17995	17737	8236.24	px^2
in_frame_fraction_r30	normalised	0.9886	1	0.0393	unitless
in_frame_fraction_r60	normalised	0.9722	1	0.077	unitless
ring_density_r30	normalised	0.0323263	0.0316752	0.00924004	px^-1
ring_density_r60	normalised	0.0296206	0.0291471	0.00715123	px^-1
roots_per_100px_perimeter	normalised	3.4497	3.3497	1.7975	per 100 px
ring_attached_length_r30_px	network v2	251.2	243	154.322	px
ring_attached_length_r60_px	network v2	427.52	404	287.022	px
ring_length_w_r30_px	network v2	316.39	292.936	182.242	px
ring_length_w_r60_px	network v2	626.271	564.245	369.011	px
sholl_crossings_r10	network v2	7.32	8	3.3131	count
sholl_crossings_r20	network v2	7.24	7	4.2257	count
sholl_crossings_r30	network v2	7.44	6	4.1138	count
loop_count	loops, report only	1.76	1	2.4542	count
loop_edge_share	loops, report only	0.1352	0.1316	0.1215	share
ring_width_mean_r30_px	width	7.6443	7.3803	0.8746	px

Per field
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measure	value	units
analysed_area_px2	945278	px^2
total_skeleton_length_px	25440	px
canalicular_length_density_per_px	0.0269127	px^-1
junction_count	493	count
junction_density_per_px2	0.00052154	px^-2
lacuna_count	29	count

lacunae_per_px2	3.068e-05	px^-2
skeleton_component_count	356	count
mean_component_length_px	71.4607	px
median_component_length_px	25.5	px
field_length_density_w_per_px	0.030574	px^-1
field_density_without_flagged_per_px	0.0269566	px^-1
field_density_in_roi_per_px		px^-1
unattached_length_px	270	px
unattached_component_count	16	count
canalicular_length_density_attached_per_px	0.0269127	px^-1
field_length_density_with_unattached_per_px	0.0271983	px^-1
loop_count	53	count
loop_edge_share	0.1509	share
edge_count	1332	count

Canalicular width, the new measure

measure	value	units
width_mean_px	7.8677	px
width_median_px	8	px
width_p10_px	4	px
width_p90_px	10.7703	px
width_pixels_used	25144	count

Canalicular width: 2 x the Euclidean distance transform of the canalicular mask at each skeleton pixel. Measured only on skeleton pixels on the canalicular mask: the lacuna buffer and the dilated vascular mask are excluded. The threads are about 2 to 6 px wide, so the width is close to the pixel grid and quantised in steps of about 1 px, it moves with the threshold (a lower cut widens every thread), and it has not been validated against any manual measurement.

Lacuna shape

lacuna_id	area_px2	major_axis_length_px	minor_axis_length_px	aspect_ratio	eccentricity	solidity	orientation_rad	centroid_row_px	centroid_col_px	on_border
1	468	33.3861	22.2869	1.498	0.7446	0.7647	0.8461	10.1432	885.363	True
2	2181	148.407	31.7971	4.6673	0.9768	0.5491	-0.2241	127.792	674.16	False
3	3003	146.448	39.2919	3.7272	0.9633	0.5868	-0.1342	219.3	634.066	False
4	4348	102.17	64.0204	1.5959	0.7793	0.7299	-0.8016	123.509	117.145	False
5	6103	130.52	74.6261	1.749	0.8204	0.7441	-0.6975	211.602	115.526	False
6	4713	113.091	66.6058	1.6979	0.8082	0.6869	0.0149	166.949	811.934	False
7	2838	120.116	56.4992	2.126	0.8825	0.5006	-0.0922	183.402	482.588	False
8	640	32.9351	28.0278	1.1751	0.5252	0.8112	-0.2136	241.733	39.7797	False
9	593	40.0782	21.7723	1.8408	0.8396	0.7583	0.4372	249.45	834.877	False
10	5444	169.703	43.0843	3.9389	0.9672	0.8492	-0.1663	315.438	989.012	False
11	472	35.2068	21.2692	1.6553	0.7969	0.7528	-0.5629	318.485	581.102	False
12	5032	145.049	50.087	2.8959	0.9385	0.6975	-0.3567	389.803	259.63	False
13	1325	56.519	31.2238	1.8101	0.8335	0.9138	-0.4181	431.483	840.339	False
14	1869	63.382	43.4291	1.4594	0.7284	0.783	-0.4231	489.012	823.251	False
15	4023	130.997	44.5634	2.9396	0.9404	0.8529	-0.2947	478.273	373.626	False
16	2063	91.1753	31.0575	2.9357	0.9402	0.8892	-0.3885	511.078	505.802	False
17	38997	474.656	126.173	3.7619	0.964	0.8704	-0.1612	856.424	971.407	True
18	2736	102.242	36.178	2.8261	0.9353	0.8653	-0.4465	632.235	931.07	False
19	2738	96.5558	37.4594	2.5776	0.9217	0.8784	-0.3793	679.917	399.446	False
20	3288	105.667	42.059	2.5123	0.9174	0.8761	-0.3815	724.683	177.336	False
21	797	46.6745	23.1748	2.014	0.868	0.8935	-0.2375	760.548	828.605	False
22	584	33.791	23.1054	1.4625	0.7297	0.9125	-0.7852	793.788	813.11	False
23	1183	68.5981	39.5227	1.7357	0.8173	0.5796	-0.4587	845.742	872.086	False
24	518	37.316	23.4561	1.5909	0.7777	0.7443	0.7533	887.49	867.025	False

25	4334	128.945	44.2583	2.9135	0.9393	0.8796	-0.1396	903.815	374.562	False
26	1983	74.3475	38.2775	1.9423	0.8573	0.8154	-0.0273	970.974	775.836	False
27	593	45.6756	20.4671	2.2317	0.894	0.7907	1.5552	1015.45	766.083	True
28								7.45	906.793	True
29								686.638	523.493	False

Network per lacuna

lacuna_id	roots_count	ring_length_r30_px	ring_length_r60_px	owned_length_px	edge_count	mean_edge_length_px	ring_width_mean_r30_px
1	6	79	179	174.238	11	15.8398	6.6408
2	21	392	645	542.245	38	14.2696	7.9442
3	14	291	498	731.097	40	18.2774	7.1502
4	8	341	587	577.387	31	18.6254	7.2773
5	10	261	583	617.286	23	26.8385	7.566
6	20	457	784	865.595	40	21.6399	7.528
7	19	723	1416	1324.53	81	16.3522	6.6187
8	5	148	269	487.375	25	19.495	6.7585
9	6	202	455	726.612	31	23.4391	7.3169
10	10	325	580	1119.45	52	21.5278	7.3803
11	5	236	415	172.51	11	15.6827	6.8321
12	20	632	1263	2681.43	133	20.1611	7.0906
13	4	90	197	79.1838	7	11.312	7.454
14	8	165	372	227.338	12	18.9448	6.9494
15	13	411	890	1092.82	61	17.9151	8.8041
16	10	291	607	967.637	42	23.039	8.2932
17	5	302	559	236.581	9	26.2867	7.0501
18	6	228	401	566.186	25	22.6474	9.0318
19	11	305	670	1297.54	48	27.0322	8.1069
20	10	349	860	1189.62	50	23.7924	7.3406
21	4	91	213	230.208	16	14.388	6.8986
22	2	94	143	75.8406	2	37.9203	8.0227
23	7	187	245	405.961	26	15.6139	6.6557
24	1	44	91	114.983	3	38.3276	9.9543
25	10	375	855	842.269	33	25.5233	8.5809
26	4	150	298	252.522	11	22.9565	6.6699
27	2	13	13	13.6569	1	13.6569	3.8495
28	3	53	74	57.799	3	19.2663	7.0183
29	5	199	414	1157.48	41	28.2312	8.8823

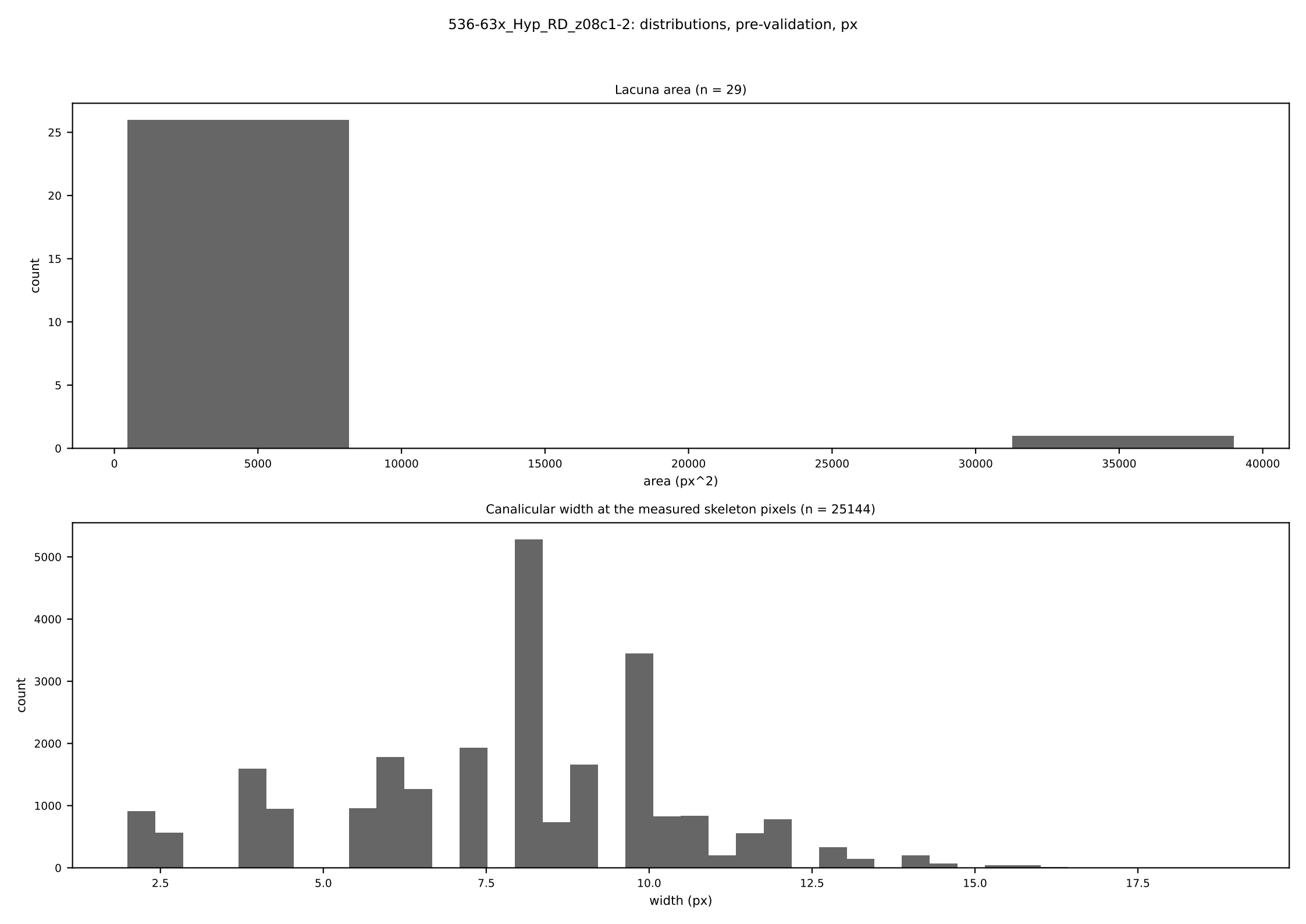
Normalised per lacuna

lacuna_id	perimeter_px	ring_area_r30_px2	ring_area_r60_px2	in_frame_fraction_r30	in_frame_fraction_r60	ring_density_r30	ring_density_r60	roots_per_100px_perimeter
1	111.598	2420	6128	0.6134	0.5737	0.0326446	0.0292102	5.3764
2	415.641	10082	20733	1	1	0.0388812	0.0311098	5.0524
3	495.973	9778	17737	1	1	0.0297607	0.0280769	2.8227
4	488.535	9556	20455	1	1	0.0356844	0.0286971	1.6375
5	609.269	10724	19977	1	0.9991	0.0243379	0.0291836	1.6413
6	442.446	12800	26204	1	1	0.0357031	0.0299191	4.5203
7	514.931	14099	31059	1	1	0.0512802	0.0455907	3.6898
8	123.439	3719	6442	0.9752	0.821	0.0397956	0.0417572	4.0506
9	128.539	4684	10910	1	1	0.0431255	0.0417049	4.6678
10	423.747	11743	23622	0.8152	0.7003	0.0276761	0.0245534	2.3599
11	117.539	5258	12354	1	1	0.044884	0.0335924	4.2539
12	487.973	14467	32198	1	1	0.0436856	0.039226	4.0986
13	156.16	5263	12719	1	1	0.0171005	0.0154886	2.5615
14	228.202	6707	15935	1	1	0.0246012	0.0233448	3.5057
15	348.433	12410	27080	1	1	0.0331185	0.0328656	3.731

16	227.38	9187	22623	1	1	0.0316752	0.0268311	4.3979
17	1282.44	13782	18371	0.4486	0.431	0.0219126	0.0304284	0.3899
18	274.735	8709	16073	1	1	0.0261798	0.0249487	2.1839
19	250.693	9821	23303	1	1	0.0310559	0.0287517	4.3878
20	298.534	10681	26849	1	1	0.0326748	0.032031	3.3497
21	131.776	4471	9529	1	1	0.0203534	0.0223528	3.0355
22	102.847	3623	7408	1	1	0.0259453	0.0193035	1.9446
23	259.557	5651	7254	1	1	0.0330915	0.0337745	2.6969
24	120.119	2651	3753	1	1	0.0165975	0.0242473	0.8325
25	340.877	12243	29334	1	0.982	0.0306298	0.0291471	2.9336
26	234.309	6488	13038	0.9258	0.8017	0.0231196	0.0228563	1.7071
27	129.012	1451	2663	0.5828	0.5538	0.00895934	0.00488171	1.5502
28	69.2132	1534	4101	0.5962	0.5598	0.0345502	0.0180444	4.3344
29	49.1127	4216	13286	1	1	0.0472011	0.0311606	10.1807

Network v2 per lacuna

lacuna_id	ring_attached_length_r30_px	ring_attached_length_r60_px	ring_length_w_r30_px	ring_length_w_r60_px	sholl_crossings_r10	sholl_crossings_r20	sholl_crossings_r30
1	79	141	94.3259	210.551	2	2	2
2	342	438	441.747	736.24	8	12	10
3	279	358	330.007	564.245	8	6	5
4	273	470	389.676	672.541	10	11	9
5	248	452	292.936	662.299	11	7	6
6	428	713	517.546	895.495	10	10	12
7	694	1052	813.127	1593.25	14	11	16
8	92	116	170.752	314.12	6	5	4
9	202	435	228.924	526.002	6	3	5
10	325	552	368.291	663.198	9	10	7
11	179	179	272.35	480.002	6	3	5
12	563	1136	709.985	1430.87	13	16	16
13	76	76	102.326	221.995	4	0	2
14	150	205	184.196	422.818	3	5	5
15	371	812	458.635	1009.29	8	11	13
16	243	396	331.321	687.713	12	7	10
17	219	301	335.735	632.5	8	7	12
18	228	336	269.108	467.103	7	8	6
19	305	602	350.978	779.181	8	9	8
20	261	536	373.551	951.796	8	11	11
21	91	195	104.083	248.622	3	3	2
22	46	57	108.397	168.409	3	4	6
23	187	244	221.137	288.907	6	4	3
24	40	58	48.799	100.184	2	1	2
25	329	642	423.019	955.595	10	15	12
26	129	224	172.51	347.919	4	4	5
27	13	13	13.6569	13.6569	0	0	0
28	53	53	57.799	81.6985	1	1	0
29	199	404	226.338	468.99	4	5	6


536-63x_Hyp_RD_z08c1-2: distributions, pre-validation, px
Lacuna area (n = 29)
count
area (px^2)
0
5
10
15
20
25
0
5000
10000
15000
20000
25000
30000
35000
40000
Canalicular width at the measured skeleton pixels (n = 25144)
count
width (px)
0
1000
2000
3000
4000
5000
2.5
5.0
7.5
10.0
12.5
15.0
17.5